TED Public Procurement Notices Classification Report

Data Cleaning

Several data cleaning steps were performed according to the recommendations of Sections 3 and 4 of the documentation (TED Notes & Codebook).

1. All canceled contracts were removed from the dataset by filtering out entries where `CANCELLED==1`, as these contracts did not proceed to completion. [Section 3.9]

2. Since a single contract award notice (CAN) can contain information about multiple contract awards (CAs), and each CA can have multiple lots, much of the data is duplicated across multiple rows. The dataset was filtered at the contract award level by filtering down to unique `ID_AWARD` values so that each awarded contract is only counted once. [Sections 3.3-3.6]

3. Many columns contained data irrelevant to the classification of “usual” and “unusual” contracts, so they were removed from the dataset. Others, namely the contract value columns, had conflicting or missing data, so they were merged based on the information given for each variable listed in the Codebook. [Section 4]

4. There were 7,849 unique Contract Procurement Vocabulary (CPV) codes. The first two digits refer to the division (rather than the exact description) of the supplies, works, or services in the subject of the contract. A new column containing just the first two digits of each CPV code was added so the contracts could be examined more broadly. The names of the CPV divisions were also added by merging the TED CSV with a CSV containing the division codes and their corresponding titles, available here: https://github.com/jeffreyhding/ted/blob/main/cpv_divisions.csv

Classification

Two methods were used to classify contracts as "usual" or "unusual."

Method 1: CPV Frequency Classification

The first method relied on CPV frequency, where contracts with rare CPV codes were marked as "unusual.” The threshold for rarity was determined by computing the mean number of contracts per CPV (roughly 211.5), and contracts with CPV codes with below-average occurrences were classified as "unusual."

This approach was chosen because CPV codes that appear infrequently tend to represent specialized procurement categories, which would have fewer competing firms and more concentrated winners among firms that did compete.

Method 2: CPV Intradivision Frequency Classification

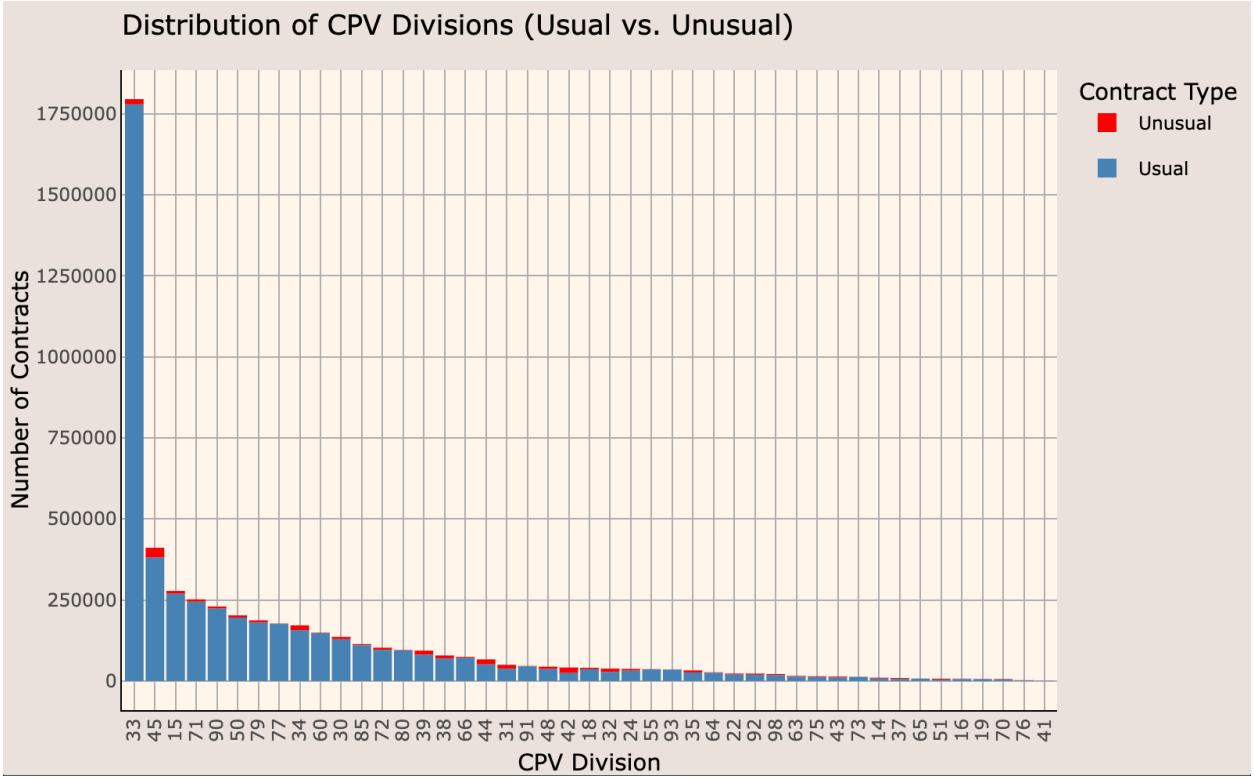
The second method focused on CPV division-based classification, which considered the relative proportion of a CPV code within its own CPV division rather than in the whole dataset. Within each division, the 10th percentile of CPV proportions was used as a threshold to identify rare CPVs. Any CPV with a proportion below its division’s threshold was classified as "unusual."

This approach accounted for the fact that some CPV divisions were much larger than others; the most niche procurement contracts in the largest divisions would have a higher overall frequency than the most common contracts of a smaller division. This method attempts to reduce bias introduced by disproportionately large CPV divisions.

Contracts were labeled “unusual” if they were deemed so by **either** of these methods.

Summary Results

Summary tables for “usual” and “unusual” classification frequencies overall and by division are available here: https://github.com/jeffreyhding/ted



I attempted another method where I tried to sort the frequencies of the `MAIN_ACTIVITY` column, which contains information on the function of expenditures (from Section 4 of documentation), but I couldn’t figure out a good way to sort the values.